

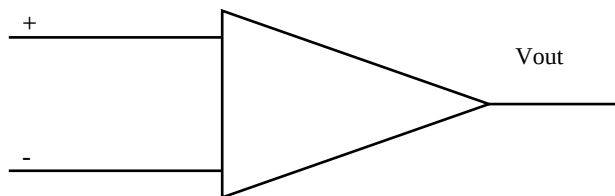
Analog Electronics – Unit 1 Complete OpAmp Notes (Exam Ready)

These notes include definitions, derivations, formulas and simplified circuit diagrams for the most important Operational Amplifier topics commonly asked in B.Tech exams.

1. Ideal OpAmp Characteristics

1. Infinite open loop gain ($A \rightarrow \infty$)
2. Infinite input impedance (no input current)
3. Zero output impedance
4. Infinite bandwidth
5. Infinite CMRR
6. Zero offset voltage

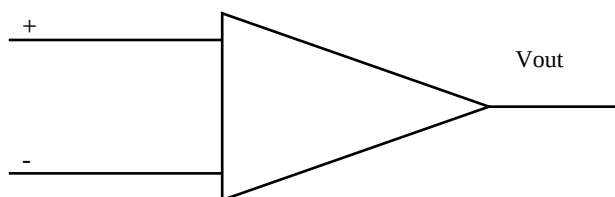
Basic OpAmp Symbol



2. Inverting Amplifier

Input signal is applied to the inverting terminal through resistor R_1 and feedback resistor R_f is connected between output and inverting terminal.

Inverting Amplifier Concept



Derivation:

Assume ideal opamp:

$V_+ = V_-$ (Virtual short)

Since noninverting terminal is grounded $\rightarrow V_- = 0$

Input current:

$$I = V_{in} / R_1$$

No current enters opamp so same current flows through feedback resistor.

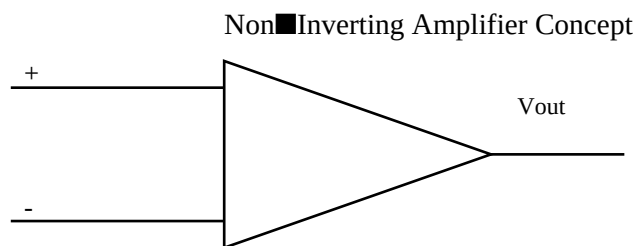
$$I = -V_{out} / R_f$$

$$V_{in} / R_1 = -V_{out} / R_f$$

Voltage Gain:

$$A_v = V_{out} / V_{in} = -R_f / R_1$$

3. Non-Inverting Amplifier



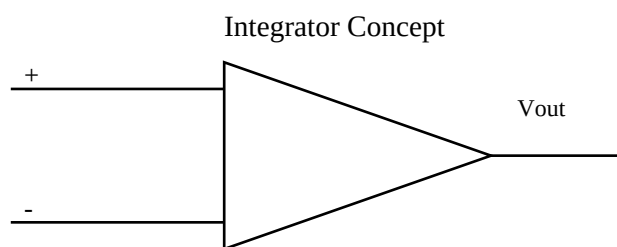
Input applied to non-inverting terminal.

Voltage divider formed by R_1 and R_f at inverting terminal.

Gain:

$$A_v = 1 + (R_f / R_1)$$

4. Op-Amp Integrator

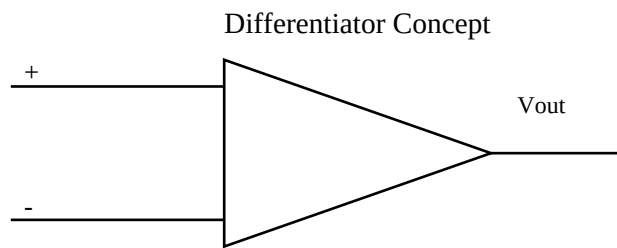


Feedback element is capacitor instead of resistor.

Output Equation:

$$V_{out} = - (1/RC) \int V_{in} dt$$

5. Op-Amp Differentiator

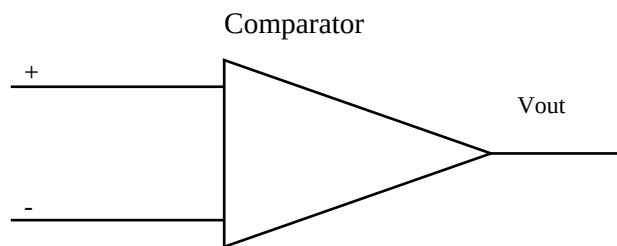


Input element is capacitor and feedback is resistor.

Output Equation:

$$V_{out} = -RC \left(\frac{dV_{in}}{dt} \right)$$

6. Comparator and Schmitt Trigger



Comparator compares input voltage with reference voltage.

If $V_{in} > V_{ref} \rightarrow$ output saturates to $+V_{sat}$

If $V_{in} < V_{ref} \rightarrow$ output saturates to $-V_{sat}$

Schmitt Trigger is comparator with positive feedback producing hysteresis.

7. Important Op■Amp Parameters

Parameter	Meaning
CMRR	Ratio of differential gain to common mode gain.
Slew Rate	Maximum rate of change of output voltage ($V/\mu s$).
Input Offset Voltage	Small input voltage required to make output zero.
Gain Bandwidth Product	Gain $\times$ bandwidth constant for op■amp.